

HDD-BIO

Discriminating Forms of Biological History Dependence

A Companion to HDD-ISA and HDD-ESA

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Status at a Glance

Element	Current status
Definitions of the five constructs and CO	Proposed
Experimental and statistical criteria (§5.1, §6, §10.1, §13)	Proposed
Internal consistency demonstration on synthetic/toy systems	Completed, by the same author (HDD-ESA, HDD-ISA)
Application to real biological data by the author	Retrospective mapping only (§6.2); no new experiment run
Independent biological validation	Not yet attempted
Independent inter-observer application (two labs applying the same checklist)	Not yet attempted

This table is the fastest way to see what the rest of this document argues in detail: a specified methodology, exercised so far only on the author's own synthetic systems and one retrospective mapping exercise, awaiting independent use.

Abstract

History-Dependent Dynamics (HDD) is a methodological framework for separating distinct claims about systems whose future behavior depends on their past. HDD distinguishes five constructs — predictive history dependence (C-I), causal trajectory dependence (C-II), feedback recurrence (C-III), functional self-reference (C-IV), and self-modeling (C-V) — governed by an Inferential Prohibition: evidence for one construct does not, by itself, authorize inference to another.

HDD-BIO is the biological companion to HDD-ISA (artificial architectures) and HDD-ESA (engineered systems). It applies the same inferential structure to living systems, from cellular networks to nervous systems and whole organisms, and adds an applicability-bounded organizational criterion (CO) drawn from the autopoiesis and organizational-closure literature.

HDD-BIO's contribution is not a new theory of biological memory, cognition, or self-modeling; it is a set of experimental reporting standards — a checklist discipline — that can be layered onto existing and future biological experiments so that predictive, causal, mechanistic, and representational claims are not silently conflated with one another. Where a construct overlaps with established programs such as predictive processing, active inference, or forward-model theory, this document names the overlap explicitly rather than implying novelty that is not present; its value lies in the discipline of separation itself, not in priority over those programs. The status-at-a-glance table above states which parts of the framework are proposed versus demonstrated; §15 specifies what would count as validating the framework itself; and provenance is addressed once, directly, in §19.1.

1. Introduction

Biological systems exhibit persistent effects of prior states at many scales: cells alter future responses after prior signal exposure; regulatory networks show hysteresis and bistability; regenerative systems retain information about past physiological states; nervous systems exploit recurrent circuitry and predictive mechanisms. These phenomena

are routinely described with overlapping vocabulary — memory, adaptation, hysteresis, history dependence, recurrence, self-reference, self-modeling, cognition — that HDD-BIO treats as related but non-equivalent.

Three levels are kept distinct throughout:

- Level A — Phenomenon: the organism responds differently depending on its history.
- Level B — Mechanism: a specified causal organization (feedback, recurrence, latent state) maintains or produces that difference.
- Level C — Interpretation: the organism possesses a self-model, functional self-reference, or organizational closure.

No inference from one level to the next is authorized without construct-specific evidence. The five HDD constructs, briefly:

- C-I — Predictive history dependence: past information improves out-of-sample prediction beyond the measured present state.
- C-II — Causal trajectory dependence: changing the prior trajectory changes the future under a matched present condition.
- C-III — Feedback recurrence: a history effect is causally mediated by a specified recurrent pathway.
- C-IV — Functional self-reference: the system causally uses a signal referring to its own state or consequences.
- C-V — Self-modeling: the system predicts its own future states, represents counterfactuals about itself, and uses those predictions to act.

The objective of HDD-BIO is not to establish that biology is history-dependent — that is already well documented. The objective is to determine which form of history dependence is present in a given experiment, what causal evidence supports it, and what further inference remains unauthorized.

2. Relation to HDD, HDD-ISA, HDD-ESA, and HDO

HDD-BIO belongs to a family of documents by the same author, organized as a tree rather than as a flat sequence. HDD (the general framework) is the root document. HDD-ISA (artificial architectures), HDD-ESA (engineered dynamical systems), and HDD-BIO (this document) are its three domain-specific companions, each translating the same constructs into a different class of system. HDO (a philosophical interpretation) is not a fourth sibling at the same level as the three domain companions: it branches specifically from HDD-BIO, as a philosophical companion to the biological case rather than to the general framework directly. The position of the HDD Ethical Framework (governance under uncertainty) within this tree is not specified here and is left an open question; it should not be assumed to branch from any single domain companion on the basis of this document. Readers should treat the whole family as one author's internally consistent research program rather than a body of independently corroborated results; cross-references among these documents establish coherence, not external validation.

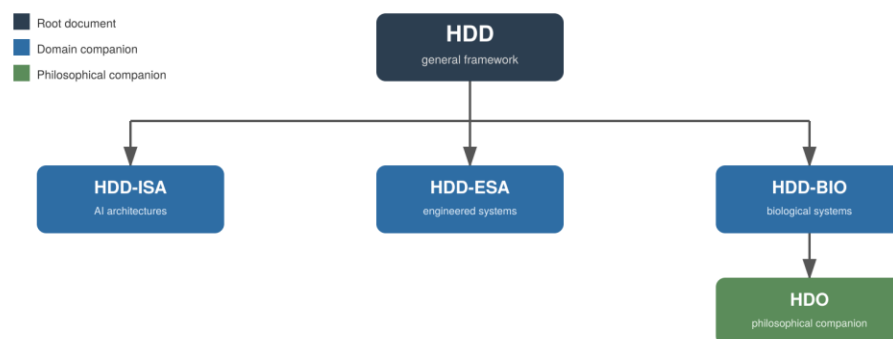


Figure 1. Document tree for the HDD project family. HDD is the root; HDD-ISA, HDD-ESA, and HDD-BIO are its domain-specific companions; HDO branches from HDD-BIO specifically. The HDD Ethical Framework is omitted from this diagram because its position in the tree relative to the domain companions has not yet been finalized.

2.1 HDD

Defines the general problem — when does a system's history make a difference to its future, and what kind of difference is it? — and separates predictive, causal, recurrent, self-referential, and self-modeling claims.

2.2 HDD-ISA and HDD-ESA

Translate the constructs into experimentally addressable interfaces for AI architectures and engineered systems respectively. Both emphasize testability over labels: whether a functional property can be causally accessed, perturbed, and discriminated, not whether a system merely contains a memory module, recurrent layer, or feedback loop by name.

2.3 HDD-BIO

Applies the same separations to living systems, where the measured state is usually incomplete, relevant variables span multiple spatial and temporal scales, and organizational closure may be meaningfully defined for some systems and not others.

2.4 HDO

HDO branches from HDD-BIO specifically, as its philosophical companion, and asks what stronger ontological interpretation may follow if historical organization becomes constitutive of a process's identity. That step is explicitly abductive, not deductive, and HDD-BIO does not import it: biological evidence gathered under this framework is not automatically an ontology of life.

3. Relation to Existing Literature on Prediction, Agency, and Self-Modeling

The distinctions HDD-BIO draws for C-IV and C-V are not without precedent, and this document does not claim to be the first to separate generic prediction from self-specific prediction; positioning them explicitly against established programs keeps them from being silently absorbed into those programs. Several existing research programs address closely related ground:

- Forward-model / internal-model theory (Wolpert, Ghahramani, Flanagan, and colleagues) already distinguishes efference copy from sensory feedback and treats corollary discharge as a predictive mechanism used to cancel self-generated sensory consequences — the same phenomenon this document cites as a C-IV candidate interface.
- Active inference and predictive-processing frameworks (Friston and colleagues) propose that organisms minimize prediction error using generative models that can include the organism's own states and actions, which overlaps substantially with the self-vs-world contrast this document requires for C-V.
- Comparative cognition research on episodic-like memory and mental time travel (Clayton and colleagues; Suddendorf and Corballis) already investigates whether non-human animals represent their own past and future states, a question closely related to V1 and V2 as defined here.
- Philosophical work on self-models (Metzinger) and on the biological grounding of selfhood (Damasio) has long argued that a self-model must be causally efficacious rather than merely correlated with self-related information — the same requirement encoded here as V3.

HDD-BIO's distinct contribution, properly stated, is narrower than a new theory of self-modeling: it is a reporting and identifiability discipline — a checklist of statistical and causal conditions (preregistered history windows, matched-present designs, residual-latent sensitivity analysis, self-vs-world controls) that can be applied on top of any of the programs above to prevent a positive result in one of them from being silently reported as evidence for a stronger construct. Where this document uses terms such as self-reference or self-modeling, it does so in the

operational sense defined in Sections 7–8, not as a claim to have resolved, or to supersede, the debates in the cited literature.

4. Observation-Model Problem

4.1 Partial observability

The complete state of a biological system is almost never experimentally available. An experiment may measure membrane potential, gene expression, protein abundance, neural activity, calcium dynamics, hormone concentration, morphology, or behavior, while the true state includes many unmeasured variables. Even if the complete state evolves as a Markov process, the measured state need not: earlier observations can carry information about latent components of the true state that is simply absent from the present measurement.

This is the central observation-model problem for HDD-BIO.

4.2 What a positive C-I result actually establishes

If a history-dependent model predicts future observations better than a present-only model, the immediate and only warranted conclusion is that the measured present state is predictively insufficient for the specified prediction task, given the chosen representation and model class. This does not by itself establish a stored memory, a recurrent mechanism, a particular latent variable, self-reference, self-modeling, or consciousness.

4.3 Mitigation

HDD-BIO recommends, where feasible: intervention rather than observation alone; multimodal measurement; explicit latent-state modeling; cross-subject and cross-session validation; preregistered history windows; and transparent reporting of the inferential bound that remains after these steps. Partial observability is not eliminated by these measures — in many biological systems it cannot be — but the remaining uncertainty is made explicit rather than absorbed silently into the conclusion.

5. C-I — Predictive History Dependence

C-I asks whether information from the past improves out-of-sample prediction beyond the measured present state and current inputs. The construct is purely predictive; it is not a semantic claim about memory.

5.0 Formal estimand

Let $Y(t+h)$ be the outcome variable at horizon h , $S(t)$ the measured present state and current inputs, and $H(t) = \{Y(t-1), Y(t-2), \dots, Y(t-k)\}$ a preregistered history window of length k . Let M_{present} be a model fit to predict $Y(t+h)$ from $S(t)$ alone, and M_{history} a model of matched capacity fit to predict $Y(t+h)$ from $S(t)$ and $H(t)$ jointly. Let $L(\cdot)$ be a prespecified, out-of-sample scoring rule (e.g., negative log-likelihood or mean squared error), evaluated on temporally held-out data.

The C-I estimand is:

$$\Delta = L(M_{\text{present}}) - L(M_{\text{history}})$$

with the sign convention fixed so that $\Delta > 0$ indicates the history-augmented model achieves lower loss. C-I is supported when the estimated Δ , reported with a confidence or credible interval from a block-bootstrap or permutation null appropriate to autocorrelated time series, exceeds a preregistered minimum effect size δ_{min} , under model capacities matched by parameter count or an explicit complexity penalty, and validated out-of-sample across held-out time windows and, where possible, held-out animals or sessions. This is an implementable estimand: k , h , L , δ_{min} , the capacity-matching rule, and the partitioning scheme must all be stated in the preregistration before Δ is computed, not chosen after inspecting the result.

5.1 Minimal statistical specification

Because a more expressive model can almost always improve some metric on some dataset, C-I requires a prespecified protocol. At minimum, a C-I study should fix, in advance of analysis:

- A prediction metric (log-likelihood, mean squared error, or another domain-appropriate scoring rule).
- A prediction horizon and, separately, a preregistered history window — post hoc window selection is a form of overfitting.
- Equivalent model capacity between the history-dependent and present-only models, or an explicit complexity penalty.
- Temporally separated training, validation, and test partitions, with cross-animal or cross-session validation where possible.
- A null distribution (permutation or block-bootstrap) appropriate to autocorrelated biological time series.
- A present-only model that already includes candidate latent variables, so that any remaining improvement is not merely evidence that the baseline model was under-specified.
- A preregistered minimum effect size counted as evidence, reported together with a confidence or credible interval.

5.2 Interpretation and status

A positive C-I result means history contains predictive information not captured by the measured present state; it does not mean the organism possesses memory in any stronger sense, since latent physiological variables, delayed dynamics, or incomplete state measurement can generate the same statistical signature.

Status. Within the author's own synthetic demonstrations (HDD-ESA, HDD-ISA), where the complete state is known by construction and the observation model is fully controlled, C-I has been shown to behave as predicted. This is an internal consistency check on toy systems designed by the same author, not an independent replication, and it does not itself constitute evidence about any biological system. Biological application remains proposed, subject to the specification in §5.1.

6. C-II — Causal Trajectory Dependence

C-II asks whether two systems that reach an appropriately matched present condition through different histories nonetheless show a different future response under a matched subsequent intervention. C-I establishes predictive relevance; C-II manipulates history directly, attempting to distinguish history as predictor from history as causal determinant.

6.0 Formal estimand

Let H denote a history-manipulation variable taking values $H1$ (control trajectory) and $H2$ (perturbed trajectory), P the set of variables required to match the present state across groups, I a subsequent matched intervention, and Y the future outcome. Using Pearl's do-operator to denote external manipulation of H rather than passive observation of it, the C-II estimand is:

$$\tau = E[Y \mid \text{do}(H = H1), P, I] - E[Y \mid \text{do}(H = H2), P, I]$$

C-II is supported when τ is estimated to be different from zero, under a prespecified matching tolerance on P , a matched I applied identically to both arms, and — because P can only ever be a finite, measured subset of the complete present state — an explicit residual-latent sensitivity analysis (§6.3) bounding how much of τ could instead be explained by unmeasured components of the present state rather than by H itself. Reporting τ without that bound is reporting a difference, not a causal trajectory effect in the HDD-BIO sense.

6.1 The planarian bioelectric paradigm as a candidate interface

Planarian regeneration is an unusually informative candidate system. Transient perturbation of endogenous bioelectric networks has been shown to produce persistent changes in regenerative target morphology that propagate across subsequent regeneration rounds, and separate work has characterized stable and stochastic pattern memories

in the same system. A related but distinct literature shows that environmental familiarization can persist across decapitation and head regeneration. HDD-BIO treats both as candidate substrates for different questions, not as a single phenomenon.

These are candidate experimental interfaces for C-II, not completed validations: the original studies were not designed around a matched-present causal estimand in the HDD sense. The translation offered here is methodological — it maps selected phenomena from the bioelectricity literature onto this document's causal vocabulary without attributing that classification to the original authors.

6.2 Worked example: retrospective mapping onto a published study

This subsection performs a retrospective mapping exercise, as invited by §15's first validation test. It does not claim that the cited study demonstrates C-I or C-II; it shows step by step what applying the checklist to an existing, real dataset would require, and where the published design already meets the checklist versus where it does not.

Shomrat and Levin (2013) trained planarians (*Dugesia japonica*) in an automated environmental-familiarization paradigm: one group was habituated to a rough-floored testing environment, a second to a smooth-floored control environment, over repeated trials, before the animals were decapitated and allowed to regenerate a new head. Regenerated animals were then retested and showed savings — faster reacquisition of the trained response than untrained controls — consistent with retention of the pre-decapitation training across head regeneration.

Mapped onto the HDD-BIO schema, without reinterpreting the original authors' own claims:

- Data: trial-by-trial response latencies/accuracy from the automated training apparatus, before and after decapitation and regeneration.
- Measured present state (S): the regenerated animal's response behavior at first retest, with no direct measurement of its internal molecular or bioelectric state.
- History (H): whether the animal underwent rough-floor or smooth-floor familiarization prior to decapitation — the manipulated trajectory variable.
- Matching (P): both groups underwent the same decapitation procedure and the same regeneration interval, which matches gross present-state variables (body size, regeneration stage) but does not measure or match finer physiological or bioelectric state.
- Analysis available for a C-I mapping: whether trial history before decapitation improves prediction of post-regeneration performance beyond present-only variables (regeneration stage, body size) — this comparison is not reported in the original paper in the Δ -and-preregistered-window form §5 requires, so a C-I classification cannot be assigned from the published data as it stands.
- Analysis available for a C-II mapping: the reported savings effect is a difference between matched-present groups with different histories (rough-floor vs. smooth-floor training) — structurally aligned with the C-II estimand in §6.0 — but the original design does not report a residual-latent sensitivity bound (§6.3), so τ cannot yet be separated from an unmeasured physiological difference between groups.

Result of the mapping exercise: this published study is a strong structural candidate for both C-I and C-II, and illustrates exactly what §6.0's estimand and §6.3's sensitivity requirement would ask a future study (by this or any other author) to add — a preregistered history window and scoring rule for C-I, and an explicit residual-latent bound for C-II. Under the categories in §13.1–§13.2, the correct classification of the existing published result is not demonstrated for C-I (the necessary statistical comparison was not run) and underdetermined for C-II (the causal contrast is structurally present but the sensitivity analysis is absent), not negative and not validated. This is the intended output of applying the checklist retrospectively: a precise statement of what is missing, not a verdict on the original study's quality, which was not designed against this checklist in the first place.

6.3 Residual-latent sensitivity

Even when the measured present states of two history groups are matched, unmeasured components of the true state may still differ between them, and a future difference in outcome would then be explained by that residual latent-state mismatch rather than by the trajectory itself. A C-II study should therefore report explicitly: which latent variables were measured; which remain unmeasured; which auxiliary predictions would distinguish a trajectory

effect from a residual-latent effect; and what magnitude of unmeasured difference could account for the observed outcome difference. This is a sensitivity analysis, not a claim that latent confounds can be eliminated — in most biological systems they cannot be.

Status: proposed. The planarian bioelectric-memory system is a strong candidate paradigm, not a completed HDD validation. Negative C-II results require the identifiability conditions in §11 and should generally be reported as not demonstrated or underdetermined rather than negative.

7. C-III — Feedback Recurrence

C-III asks whether a history-dependent effect is causally mediated by a specified, temporally functional feedback pathway. Biological systems contain cycles almost everywhere; a causal graph containing a cycle is not the same as a temporally functional cycle whose selective disruption alters the specific history-dependent effect under investigation. Support requires: the history effect exists under intact feedback; selective disruption of the candidate pathway substantially reduces or eliminates it; the intervention is sufficiently specific; a control pathway does not produce the same elimination; and general viability remains adequate for interpretation (destroying the whole system trivially removes every effect and establishes nothing about mediation).

7.1 Perturb-seq as a discovery interface, not a validation

Perturb-seq studies combining CRISPR perturbation with single-cell sequencing have identified candidate regulatory structures — for example, an isolated feedback loop involving the translocon and IRE1 α in the unfolded protein response — and independent biochemical work has shown that the Sec61 translocon physically constrains IRE1 α activation and attenuation, with selective knockdown and point mutants producing correspondingly altered kinetics. This is genuine causal manipulation of the candidate pathway and strengthens it as a C-III candidate, but it targets a steady-state regulatory relationship, not a previously established history-dependent effect: no prior trajectory is manipulated and no historical effect is shown to be reduced. A strict C-III test would still need to begin from a demonstrated C-I or C-II effect involving this pathway and then show that disrupting it — against a capacity-matched sham — selectively reduces that historical effect.

A practical caution for reuse of public Perturb-seq atlases: a dataset's accessibility and its coverage of a specific pathway are independent properties. A screen built around a different selection criterion (for example, a growth-based genetic-interaction panel) should not be substituted for a screen specifically targeting the pathway of interest merely because it is more widely distributed.

Status: proposed. Candidate feedback structures can be identified with existing methods, but C-III requires selective causal perturbation of a temporally functional cycle connected to a previously established history-dependent effect.

8. C-IV — Functional Self-Reference

C-IV asks whether the system causally uses a signal referring to its own state, activity, or consequences. A variable correlated with the organism's own state is not automatically a self-representation; the signal must enter the causal organization of the system in a way that affects function when appropriately perturbed. HDD-BIO operationalizes this with three jointly required conditions:

- IV-1 Self-specificity: the candidate signal carries information about the system's own state or action that is not explained by external stimulation alone.
- IV-2 Functional dependence: disrupting downstream use of the signal, while preserving the signal itself, measurably alters function.
- IV-3 Counterfactual specificity: selectively disrupting the self-referential channel alters predicted function, while disrupting a matched non-self channel under otherwise identical conditions does not.

Corollary-discharge and efference-copy signals — well characterized across many animal systems as forward-model mechanisms that support discrimination between self-generated and externally generated sensory input — and human auditory paradigms showing suppressed responses to self-generated sounds are strong candidate interfaces

for IV-1 and IV-2. As noted in §3, these mechanisms are already studied extensively within forward-model theory; what C-IV adds is the requirement that IV-3 be tested explicitly, since IV-1 and IV-2 alone do not separate a self-specific channel from generic predictive processing.

A corollary discharge can support prediction of sensory consequences without requiring a temporally extended model of the organism itself, so C-IV does not, by itself, establish C-V. Molecular self/non-self discrimination (e.g., immune recognition) is likewise not automatically C-IV under this operational definition, since it need not satisfy IV-2 or IV-3 as causal-use conditions.

Status: proposed, subject to $IV-1 \wedge IV-2 \wedge IV-3$.

9. C-V — Self-Modeling

C-V is the most demanding construct: does the system predict its own future states (V1), represent counterfactual alternatives involving itself (V2), and causally use those predictions to select actions (V3)? Generic environmental prediction is insufficient; the question is whether the organism predicts itself as part of the state-space used for action selection, not merely whether it predicts at all.

9.1 The self-vs-world contrast

A central problem is that a self-model must be identified independently of behavioral success, or adaptive behavior in any sufficiently complex controller could be read as self-modeling. C-V therefore requires a selective intervention that perturbs information about the system's own state while leaving non-self prediction intact, with the predicted pattern being: environmental prediction remains preserved; prediction of the consequences of the system's own actions is selectively impaired; and decisions depending on that second capacity are selectively altered.

Prefrontal perturbation studies of counterfactual thinking in humans (for example, work showing that anodal tDCS over left dorsolateral prefrontal cortex reduces counterfactual thinking and blunts psychophysiological reactivity to choice outcomes) provide a candidate interface for V2, and comparative-cognition paradigms on episodic-like memory and future planning provide candidates for V1. Neither line of work, on its own, demonstrates that a self-model is causally used to select policy — that is exactly the V3 condition, and it is the decisive addition this document requires beyond what those literatures already show.

9.2 Planarians and C-V

A persistent bioelectric pattern may encode a target morphology and guide regeneration without demonstrating that the organism constructs a counterfactual self-model and uses it to select actions. Pattern memory is therefore not classified as C-V here; this is exactly the category distinction the Inferential Prohibition is meant to protect.

Status: proposed, subject to $V1 \wedge V2 \wedge V3$ with a self-vs-world control. Because this criterion is stringent, a negative C-V result should be rare and reserved for designs meeting the identifiability conditions in §11; most non-positive outcomes should be reported as not demonstrated or underdetermined.

10. CO — Criterion of Organizational Coherence

CO is defined below as a concrete stepwise procedure, rather than through process/constraint notation alone, so that the criterion is directly executable rather than purely descriptive.

CO is not a sixth HDD construct in the C-I-C-V sequence; it is a separate, applicability-bounded organizational criterion drawn from the autopoiesis and organizational-closure literature (Maturana and Varela; Mossio and Moreno). Its question: when the dependencies relevant to a system's self-production are jointly disrupted, does the system's capacity for organizational self-maintenance cease, in a way that a matched non-organizational disruption does not also produce?

CO is not a consciousness test, not a self-model test, and not simply a feedback test. It applies only to systems in which organizational self-production or autopoietic closure can be meaningfully defined and experimentally probed; if that interface does not apply, the correct classification is inapplicable, not failed.

CO is not intended as a general test applicable to all biological systems, and it should not be treated as a universal property that the framework simply goes looking for. Applicability varies by system:

- Clearly applicable: a single free-living cell or a minimal synthetic protocell, where a bounded set of membrane, metabolic, and regulatory dependencies can be enumerated and jointly perturbed against a severity-matched control.
- Controversially applicable: a multicellular tissue or whole organism, where self-production dependencies exist but are distributed across many redundant and compensating pathways, making step 3 of §10.1 (enumerating the closure loop) genuinely contestable and dependent on where the observer draws the organizational boundary.
- Clearly inapplicable: an isolated molecular pathway, a single neuron, or any subsystem that does not itself reproduce the constraints that enable it — CO should not be applied at this scale merely because a feedback loop is present, since a feedback loop is not the same thing as organizational closure.

10.1 A concrete CO protocol

A CO study should proceed through the following steps, in order, before any joint perturbation is performed:

1. Define the organizational boundary: what counts as inside versus outside the system under study.
2. Enumerate the components and constraints whose continued production is candidate for closure (e.g., specific membrane structures, regulatory circuits, or metabolic cycles).
3. For each component, identify which ongoing process is hypothesized to produce or regenerate it, and which constraint that process in turn depends on — this is the closure loop being tested, stated as a small, explicit list of dependencies rather than a general claim about "the whole system."
4. Identify experimentally accessible perturbation points for each dependency in that list.
5. Design a joint perturbation targeting the specific dependencies enumerated in step 3 — disabling one component at a time (as in C-III) is not a CO test; the perturbation must target the organizationally relevant set jointly.
6. Design a matched control perturbation of equivalent severity that damages general viability (toxicity, starvation, global energy loss) without targeting the specific closure dependencies identified in step 3.
7. Apply both perturbations and compare the resulting trajectories: CO is supported only if the targeted perturbation produces loss of self-maintenance that the severity-matched general-damage control does not.
8. Classify the result as CO-positive, CO-negative (applicable test, predicted loss occurs or fails as prespecified), CO-inapplicable (the organizational interface could not be meaningfully defined for this system), or CO-underdetermined (the design could not discriminate the two perturbation conditions).

This sequence keeps the criterion of organizational necessity separable from component necessity (disabling any single pathway) and from mere global damage, which is the safeguard the stepwise procedure is designed to make explicit.

Status: applicable only where the organizational self-production interface has been explicitly defined per steps 1–3 and is experimentally testable. CO does not establish consciousness or moral status even when CO-positive.

11. Inferential Prohibition

The central rule is a set of pairwise non-implications: evidence for C-I does not establish C-II; C-II does not establish C-III; C-III does not establish C-IV; C-IV does not establish C-V; and C-V does not establish consciousness. More generally, for any two constructs, one does not imply the other in the absence of additional, construct-specific evidence. This does not mean the constructs are unrelated — they may coexist, and causal relations between them may hold in some systems — but coexistence does not remove the need for independent evidence.

Two dimensions should be kept separate when reporting a result under this framework, and neither substitutes for the other. The construct dimension names which of C-I, C-II, C-III, C-IV, C-V, or CO is being tested. The

epistemic-status dimension, defined fully in §13.1–§13.2, states how the test came out: demonstrated, not demonstrated, negative under an applicable test, inapplicable, or underdetermined, together with an identifiability rating. A result is therefore always reported as a (construct, status) pair — for example, "C-II, underdetermined" — never as a bare status without a construct or a bare construct without a status.

Construct	Positive result establishes	Does not establish	Main confound
C-I	History improves out-of-sample prediction	A memory mechanism	Latent state
C-II	Matched trajectories diverge causally	A specific feedback mechanism	Residual latent-state mismatch
C-III	Selective feedback disruption removes the history effect	Self-reference	Pleiotropy / global damage
C-IV	A self-specific channel is causally used	A self-model	Mere correlation with self-state
C-V	$V1 \wedge V2 \wedge V3$ with a self-vs-world control	Consciousness	A generic predictive controller
CO	Organizational closure disrupted as predicted	Consciousness or moral status	Global system damage

12. Common Experimental Architecture

HDD-BIO uses a common schema across constructs: system, observation, intervention, mechanistic hypothesis, outcome. The table below is a protocol map — it summarizes the required contrast and inference for each construct — not a claim that any listed biological system has already passed the corresponding test.

Construct	Typical scale	Primary contrast	Required inference
C-I	All scales	Present-only vs. history-included model	Predictive insufficiency of the measured present
C-II	Tissue, neural, organismal	Different trajectories, matched present	Causal trajectory effect
C-III	Cellular, neural	Feedback intact vs. selectively disrupted	Causal mediation by a functional cycle
C-IV	Neural, sensorimotor	Self-channel intact vs. disrupted	$IV-1 \wedge IV-2 \wedge IV-3$
C-V	Organismal	Self-model intact vs. disrupted	$V1 \wedge V2 \wedge V3$
CO	Cellular, organismal	Closure-relevant vs. severity-matched disruption	Effect on organizational self-production

13. Experimental Reporting Requirements

Every HDD-BIO study should report, at minimum: the measured state definition; the history window and its preregistration; the intervention applied; the matching criterion at the comparison point; the outcome definition; the

mechanistic hypothesis and its control condition; the remaining inferential bound; and an explicit applicability/identifiability classification for the construct under test.

13.1 Applicability categories

- Demonstrated — the test passed the prespecified criteria.
- Not demonstrated — evidence is insufficient for a positive result.
- Negative under an applicable test — the phenomenon was tested under an identifiable design and the prediction was contradicted.
- Inapplicable — the required interface does not meaningfully apply to this system.
- Underdetermined — the experiment cannot discriminate the candidate hypotheses.

13.2 Identifiability categories

- Identifiable — the design can, in principle, distinguish the construct from its alternatives.
- Non-identifiable — no experiment of that architecture can make the distinction, regardless of sample size.
- Conditionally identifiable — identifiability depends on stated auxiliary assumptions.

Failure to demonstrate is not the same as absence; inapplicable is not the same as failed; and a negative classification requires an identifiable design, with the burden of showing identifiability held especially high for C-II, C-IV, C-V, and CO.

14. Biological Scale and Cross-Scale Caution

HDD-BIO is intentionally substrate-general within biology: the same formal construct may appear at cellular (gene-regulatory or bioelectric history), multicellular (developmental or regenerative pattern memory), neural (recurrent activity, corollary discharge), or organismal (behavioral trajectories, physiological regulation) scale. The same word should not be assumed to denote the same mechanism at every scale — cellular memory is not an organismic self-model, and bioelectric pattern memory is not subjective memory. HDD-BIO exists partly to prevent this cross-scale collapse.

15. Validation of the Framework

The rest of this document specifies what a biological experiment must do to support a given construct. This section specifies, symmetrically, what would have to happen for HDD-BIO itself — the checklist, not any single experiment — to be considered a useful methodology.

HDD-BIO is itself a methodological hypothesis: the hypothesis that separating C-I through C-V and CO, and requiring the estimands and sensitivity analyses specified in §5–§10, yields distinctions that are not obtained from conventional analysis of the same data. That hypothesis has not been tested against independent use, and its usefulness should be judged by whether it is, not by the internal coherence of its definitions alone. Three tests would constitute evidence one way or the other:

15.1 Retrospective application

Applying the C-I/C-II/CO checklists to already-published biological datasets, as done for a single case in §6.2, and recording where the published design already satisfies the checklist, where it falls short, and whether the missing pieces (a preregistered window, a residual-latent bound, a severity-matched control) are things the original authors could in principle report from data they already collected. A framework that only ever finds every published study wanting, with nothing recoverable, would itself be evidence against its practical usefulness.

15.2 Prospective application

Designing and running a new experiment against the §5.0/§6.0 estimands and the §10.1 protocol, preregistered before data collection, and reporting whether the resulting (construct, status) classification changes the conclusion a conventional analysis of the same data would have reported. This is the test the author has not yet performed for any biological system; §6.2's mapping exercise is a substitute for it only in the limited sense of showing the checklist is applicable to real designs, not in the sense of showing it changes a real conclusion.

15.3 Independent replication and inter-observer application

Two or more research groups, independent of this document's author, applying the same checklist to the same or comparable data and reaching the same (construct, status) classification. This is the strongest test and the one most analogous to ordinary methodological validation in other fields (e.g., inter-rater reliability for a diagnostic instrument). Until this has happened at least once, HDD-BIO should be described as internally consistent and applicable in principle, not as validated as a methodology.

None of §15.1–§15.3 has yet been completed beyond the single retrospective mapping in §6.2. Stating this directly is itself part of what the framework asks of any experimental report (§13), and it would be inconsistent to exempt the framework's own validation status from that same requirement.

16. Planarian Bioelectric Pattern Memory as a Second Worked Example

The planarian bioelectric-memory system is valuable because it separates variables that are normally confounded: a transient bioelectric perturbation can alter future regenerative target morphology, the altered pattern can persist after the manipulation ends, and the resulting phenotype can be expressed during subsequent regeneration without repeated treatment. Unlike the behavioral-savings study mapped in detail in §6.2, this pattern-memory phenomenon has not been mapped step by step here against real reported effect sizes; it is included as a conceptual illustration of how a single system can raise several distinct HDD-BIO questions rather than one. The relevant methodological question is not whether this proves memory, but whether a transient historical intervention can alter the future space of regenerative outcomes after the original intervention has ended (a C-II-aligned question), which then opens separate questions: whether the historical information improves prediction beyond the measured present (C-I), which pathway causally maintains the effect (C-III), whether the system functionally uses a representation of its own state (C-IV), and whether it constructs and uses counterfactual self-predictions for action selection (C-V). The same biological phenomenon can serve as a platform for several of these questions without making the questions equivalent to one another.

17. What HDD-BIO Does Not Claim

HDD-BIO does not claim that all biological memory is C-I; that every hysteretic system possesses a memory mechanism; that bioelectric pattern memory establishes consciousness; that regeneration demonstrates self-modeling; that recurrence establishes self-reference; that self-reference establishes consciousness; that autopoiesis or organizational closure establishes consciousness or moral status; or that existing biological experiments have already validated all HDD constructs. The constructs are inferentially non-transitive and evidentially non-sufficient for one another — this is a claim about inference, not a claim of statistical independence.

18. Ethical Considerations

Because some experiments under this framework involve living organisms and potentially complex biological systems, the companion HDD Ethical Framework treats uncertainty itself as epistemically relevant, using the calibrated vocabulary defined here (demonstrated, not demonstrated, negative, inapplicable, underdetermined, identifiable, non-identifiable). In particular: CO does not establish moral status, and C-V does not establish consciousness. If an experiment establishes C-I but does not test C-IV, the ethical interpretation should not silently import C-IV; if CO is inapplicable it should not be reported as failed; if C-V is underdetermined it should not be reported as absent. Biological interventions can themselves alter, destroy, or reorganize the very organization under investigation, which is a further reason for conservative reporting.

19. Limitations

- Complete state matching is generally impossible, which makes C-II one of the most demanding tests and requires explicit residual-latent sensitivity analysis (§6.3).
- Latent-state confounding is reduced, not eliminated, by multimodal measurement.
- Biological mechanisms are distributed, so selective intervention can be difficult to interpret when pathways compensate for one another.
- C-IV is difficult to operationalize: correlation between a self-generated action and a neural signal is insufficient, and the $IV-1 \wedge IV-2 \wedge IV-3$ conditions raise the operational bar accordingly.
- C-V is substantially more demanding still, since generic prediction must be distinguished from self-specific, counterfactual, and causally used prediction.
- CO depends on an operational account of organizational closure; different definitions in the literature may produce different experimental interfaces even under the protocol in §10.1.
- Cross-scale equivalence cannot be assumed: the same statistical signature can arise from different biological mechanisms at different scales.
- Identifiability is not automatic — for several constructs, no experiment of a given architecture may distinguish the target hypothesis from its alternatives regardless of sample size.

19.1 Epistemic status and provenance

This document, and the HDD, HDD-ISA, HDD-ESA, HDD Ethical Framework, and HDO documents it references, are authored by a single independent researcher and self-published without peer review. Cross-references among them establish internal coherence, not external validation. Statements in earlier drafts that a construct was "validated" referred only to the author's own demonstrations on synthetic toy systems designed by the same author; those are internal consistency checks, not independent replications, and are named here to keep that distinction clear.

None of this bears on whether the underlying distinctions are correct. The value of a reporting discipline does not depend on the credentials of its proposer; it depends on whether the discipline is useful when applied to real data by researchers who did not write it. That is an empirical question about adoption and utility, not a question this document can settle about itself — it is the same question posed formally in §15. Readers are invited to test the discipline directly — apply the §5.1 specification to a C-I claim already in hand, or the §10.1 protocol to a system where organizational closure is arguable — and judge it on whether the distinctions hold up in that use, rather than on the manner of its publication.

20. Conclusion

HDD-BIO extends History-Dependent Dynamics into biological systems without collapsing biological history dependence into a single concept. It defines five constructs (C-I predictive history dependence, C-II causal trajectory dependence, C-III feedback recurrence, C-IV functional self-reference, C-V self-modeling) and one applicability-bounded organizational criterion (CO), governed by two safeguards: inapplicable is not failed, and evidence for one construct does not authorize inference to another.

Its purpose is to impose causal and inferential separation on phenomena frequently described with overlapping terms — memory, adaptation, hysteresis, recurrence, self-reference, self-modeling, and consciousness — and to make that separation checkable rather than rhetorical: each construct carries a stated positive result, a stated confound, a concrete estimand (§5.0, §6.0), and a stated inferential bound (§11), and CO carries a protocol that distinguishes organizational necessity from global damage (§10.1). Where a construct overlaps with an established research program, this document says so rather than claiming priority it does not have (§3); where the document's own status is relevant to how much weight a claim should carry, that is addressed once, directly, in §19.1, and what would change that status is specified in §15, rather than qualifying every section. The claim this document defends, stated without hedging, is this: different meanings of biological history dependence require different evidence, and an experiment should say plainly which meaning it supports and which it does not.

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Appendix A: Glossary

Term	Definition
C-I	Predictive history dependence. Historical information improves out-of-sample prediction beyond the measured present state. Epistemically relative to the observation model.
C-II	Causal trajectory dependence. The prior trajectory causally alters the future under a matched present condition, subject to residual-latent sensitivity analysis.
C-III	Feedback recurrence. A history effect is causally mediated by a specified, temporally functional feedback cycle.
C-IV	Functional self-reference. The system causally uses a signal referring to its own state or consequences (IV-1 $\wedge$ IV-2 $\wedge$ IV-3).
C-V	Self-modeling. The system predicts its own future (V1), represents self-involving counterfactuals (V2), and causally uses those predictions in action selection (V3).
CO	Criterion of Organizational Coherence. An applicability-bounded organizational test based on closure of constraint-generating dependencies (protocol in §10.1).
Identifiability	Whether an experimental architecture can, in principle, distinguish a target hypothesis from its alternatives, independent of data sufficiency.
Inferential Prohibition	The rule that evidence for one construct does not authorize inference to another, absent additional construct-specific evidence.
Observation model	The specification of which variables are measured and how they relate to the complete system state.
Residual-latent sensitivity	The C-II analysis of what magnitude of unmeasured present-state difference could account for an observed future difference.